

Binary Tree Sorting

This program adds the ability to sort the nodes and output their values. I reused the binary tree code I created for the last assignment, and after some research into tree sort, I realized that I only needed to slightly change the behavior of the toString function in the node class in order to get a properly sorted array. My previous code created the returnString by adding the current nodes value, then getting the value of all the children. I didn't get a screenshot of the previous code before I changed it, but here is a recreation with pseudocode.

```
String toString(){  
  
    String returnString = currentNodeValue;  
  
    If( lesserNode != null){  
  
        returnString += lesserNode->toString();  
  
    }  
  
    If( greaterNode != null){  
  
        returnString += greaterNode->toString();  
  
    }  
  
    Return returnString;  
  
};
```

This created the behavior of printing the tree top-down. However, with one change, I can print it out sorted. Instead of adding the current nodes value at the beginning, I add the lesser nodes first, then the current node, finally the greater nodes. This creates a recursive function which returns a string with all the sorted values.

```
// BEGIN NEW TOSTRING FUNCTION //

// This function prints the content of the node plus the contents of the children recursively
std::string Node::toString()
{
    std::string returnString = "";

    // Get all values LESS than current node

    if (getLesserChild() != nullptr)
    {
        returnString += getLesserChild()->toString();
    }

    // Add current node to string

    returnString += std::to_string(getData()) + " ";

    // Get all values GREATER than current node

    if (getGreaterChild() != nullptr)
    {
        returnString += getGreaterChild()->toString();
    }

    return returnString;
}

// END NEW TOSTRING FUNCTION
```

This is the output of the new toString function.

```
Binary Tree Test
Select a function
1. Add new value to tree
2. Search for value in tree
3. Print contents of tree
4. Exit program
Desired function: 3
Printing entire tree sorted:
1 2 3 4 6 7 9 10
```

And with that, the assignment is complete.